

P-UNIFORM

Tier 2a :: p-Hacking / Selective Reporting

At a glance: p-uniform is a meta-analytic method that estimates the true underlying effect size in a literature and tests for publication bias by modelling the full conditional distribution of significant p-values under a specific assumed effect. Unlike p-curve, it produces a bias-corrected effect size estimate alongside an evidential value test. It requires only the test statistics from significant results — no raw data needed.

1. Overview

p-uniform was developed by Robbie van Aert, Marcel van Assen, and Jelte Wicherts, with initial versions published around 2015. It belongs to the same family as p-curve — both methods work exclusively with the significant portion of a literature's p-values — but it takes a fundamentally different approach to what it tries to estimate.

Where p-curve asks 'does this set of results contain genuine evidential value, and if so, how much statistical power do the studies have?', p-uniform asks 'what true effect size would have generated exactly this distribution of significant p-values?' It inverts the selection model: given that we know only significant results were published, and given that we can model the probability of achieving significance as a function of the true effect, we can back-calculate the effect size that makes the observed p-value distribution most probable.

This makes p-uniform particularly valuable in two situations: when a researcher wants a bias-corrected effect size estimate from a biased literature, and when the heterogeneity assumptions required by other bias-correction methods (trim-and-fill, PET-PEESE) are poorly met. p-uniform* (the asterisk version, van Aert & van Assen 2021) extends the original method to handle heterogeneity directly, making it one of the most statistically sophisticated tools in the publication bias toolkit.

2. Methodological Overview

The core insight of p-uniform is that if a true effect δ exists and studies are conducted with varying sample sizes, the distribution of significant p-values is not uniform — it is concentrated toward smaller p-values for studies with higher power, and toward larger p-values (just below .05) for studies with lower power. Knowing this expected distribution, and comparing it to what is actually observed, allows both a test of whether $\delta = 0$ and an estimate of the most likely non-zero δ .

The uniform quantile transformation

For each study i with sample size n_i and observed test statistic t_i , p-uniform computes the conditional p-value: the probability of observing a test statistic at least as extreme as t_i , given that the study was significant ($p < .05$) and given a specific true effect δ . Under the correct effect size, these conditional p-values should be uniformly distributed between 0 and 1.

$$q_i(\delta) = P(T \geq t_i \mid p < .05, \delta) / P(p < .05 \mid \delta)$$

The conditional p-value for study i . Under the true effect δ , $q_i(\delta) \sim \text{Uniform}(0,1)$. Deviation from uniformity indicates either a wrong assumed effect or publication bias.

$$H_0 \text{ test: } q_i(\delta=0) \sim \text{Uniform}(0,1)$$

The publication bias / evidential value test: if results are consistent with $\delta = 0$ (no true effect), the conditional p-values under $\delta = 0$ should be uniform. Right-skew rejects H_0 , indicating evidential value.

$$\text{Estimate } \hat{\delta}: \text{ find } \delta \text{ such that } \text{median}(q_i(\delta)) = 0.5$$

The effect size estimate: the value of δ that makes the observed conditional p-value distribution most uniform, i.e., closest to $\text{Uniform}(0,1)$. This is the method-of-moments estimator.

The test statistic is computed using a sum of log-transformed conditional p-values (analogous to Fisher's method), which follows a chi-squared distribution under the null. The estimator is solved numerically — the R package puniform implements this efficiently for t , F , and r statistics.

p-uniform* — the heterogeneity extension

The original p-uniform assumes a fixed true effect (all studies estimate the same δ). When true effects are heterogeneous across studies — different populations, interventions, or contexts — the conditional p-value distribution is no longer uniform even under the correct average effect, because variance in δ creates variance in the conditional distribution.

p-uniform* (van Aert & van Assen 2021) incorporates between-study variance τ^2 directly into the selection model, estimating both the mean effect μ and the variance τ^2 simultaneously. This makes it applicable to heterogeneous literatures where the fixed-effect assumption of original p-uniform would be violated, at the cost of requiring more studies for stable estimation. Under homogeneity, p-uniform* collapses to p-uniform.

3. Example Application: Ego Depletion Revisited

Carter et al. (2015) applied both PET-PEESE and p-curve to the ego depletion literature — the widely cited finding that exerting self-control depletes a limited cognitive resource. p-uniform, applied to the same set of studies, produced a bias-corrected effect size estimate of approximately $d = 0.21$, compared to the raw meta-analytic estimate of $d = 0.62$.

What makes this example instructive is the comparison across methods: PET-PEESE estimated $d \approx 0.24$, p-curve estimated average power of approximately 31% (implying $d \approx 0.20$ – 0.25), and p-uniform landed at $d \approx 0.21$. The convergence of three different bias-correction methods on a similar, much lower estimate than the published meta-analysis is strong evidence that the literature is substantially inflated — and that these tools, despite their different mathematical foundations, are picking up the same underlying signal.

The subsequent 23-lab pre-registered replication (Hagger et al. 2016) found $d \approx 0.04$, essentially indistinguishable from zero — suggesting all three bias-corrected estimates still overcorrected in the positive direction, but were far closer to the truth than the uncorrected meta-analytic estimate of $d = 0.62$.

4. Requirements & Applicability

p-uniform shares most of its data requirements with p-curve but adds some additional constraints related to its parametric assumptions.

Test statistics and df	The exact test statistic (t , F , z , r) and degrees of freedom for each study's focal result. p-values alone are insufficient — they cannot be back-transformed to the conditional distribution without the underlying statistic and df.
Significant results only	Like p-curve, p-uniform is computed using only significant findings ($p < .05$). Non-significant results are excluded from the estimation. This is by design — the method models the selection mechanism that produced only significant results.
One test per study	A single focal test per study, corresponding to the primary theoretical claim. Averaging or aggregating multiple tests within a study is not supported and will distort the conditional distribution.
Consistent effect size metric	All studies must be estimating the same effect in the same direction. p-uniform cannot be applied to a mixed literature where some studies test effect A and others test effect B, even if nominally related. Effect size heterogeneity of the random-effects type is handled by p-uniform* but requires at least 20–30 studies.
Minimum study count	Reliable estimation requires approximately 10–15 studies for p-uniform (fixed-effect). p-uniform* requires approximately 20–30 for stable variance estimation. With fewer studies the confidence interval on δ is very wide.
Raw data required?	No. The method operates entirely on published test statistics. This is the key practical advantage shared with p-curve over multiverse and specification curve analysis.
Effect size type	The puniform R package implements p-uniform for Cohen's d (from t -tests), partial η^2 (from F -tests), and Pearson's r . Odds ratios and hazard ratios from logistic/survival models are not currently supported natively.

5. p-uniform vs. p-curve: When to Use Which

p-uniform and p-curve are close relatives that answer related but distinct questions. The choice between them depends on what you want to know.

Primary output	p-curve: evidential value test + average power estimate. p-uniform: bias-corrected effect size δ + evidential value test.
Effect size estimate	p-curve produces an effect size estimate only indirectly (via power). p-uniform produces δ directly as its primary output — this is its key advantage when a bias-corrected magnitude estimate is needed.
Heterogeneity handling	p-curve treats heterogeneity informally (it affects curve shape but is not modelled). p-uniform* models τ^2 explicitly — preferred when heterogeneity is known or expected to be substantial.
Sample size sensitivity	p-uniform is sensitive to the accuracy of reported sample sizes. p-curve is less so, because it uses the full p-value distribution rather than a parametric conditional distribution.
Recommended use	Use p-curve when diagnosing evidential value qualitatively (is there signal here?). Use p-uniform when you need a publishable bias-corrected effect size estimate and can verify the fixed-effect assumption is reasonable.
Convergence check	Best practice is to run both. If they converge on similar conclusions, confidence in the result increases substantially. Divergence signals model sensitivity and should be reported.

6. Limitations

- p-uniform assumes publication bias takes a specific form — that all studies with $p < .05$ are published and all with $p \geq .05$ are suppressed. In reality, bias is partial and probabilistic. Violation of this sharp selection assumption biases the δ estimate, typically toward overestimation of the true effect.
- The fixed-effect assumption of original p-uniform (all studies estimate the same δ) is violated in most real literatures. p-uniform* addresses this but requires more studies and has higher computational cost. Using fixed-effect p-uniform in a heterogeneous literature produces biased estimates.
- p-uniform is sensitive to focal test selection. If an analyst selects a non-focal test (e.g., a secondary outcome or a subgroup analysis) instead of the primary test, the conditional distribution is misspecified and the estimate is uninformative.
- The method cannot detect p-hacking that operates within rather than between studies — e.g., a researcher who runs many internal analyses and reports the most significant one. The published p-value is significant, which is all p-uniform sees, but it does not represent the correct comparison to the theoretical distribution.
- p-uniform has lower statistical power than p-curve for the evidential value test when the true effect is large. p-curve's right-skew test is more sensitive to the skewness signal that a large true effect produces; p-uniform's uniformity test is more sensitive to the mean effect.
- The numerical optimizer for δ can be unstable when the number of studies is small or when effect sizes are very heterogeneous. Always inspect the likelihood function and confidence interval, not just the point estimate.

7. How to Apply

R package: puniform

The puniform R package (van Aert 2022) is the standard implementation. It accepts vectors of test statistics and sample sizes and returns the evidential value test, the bias-corrected effect size estimate, its confidence interval, and the p-uniform* estimates if requested.

Minimal workflow

- Identify the literature: define inclusion criteria before examining results. Collect the focal test statistic (t, F, or r) and sample sizes for each significant finding.
- Install the package: `install.packages('puniform')` in R. Load with `library(puniform)`.

- Run the basic analysis: `puniform(tobs = c(2.3, 2.8, 3.1, ...), n = c(40, 60, 35, ...), side = 'right')` for a set of one-sided t-tests. Adjust the 'method' argument to 'P' (p-uniform) or 'LNP' (log normal p, more robust) as needed.
- For heterogeneous literatures, use the `puni_star()` function from the same package to fit p-uniform*, which estimates both μ and τ^2 .
- Run p-curve alongside using the p-curve app (pcurve.com) and compare conclusions. Report both sets of results and note any divergence.

Reporting standards

Report: the number of studies included, the focal test selection rule, the evidential value test statistic and p-value, the δ estimate with 95% confidence interval, whether fixed-effect or random-effects (p-uniform*) was used and why, and the p-curve result for comparison. If the two methods diverge, discuss which assumptions are more likely violated in this literature.

8. Related Tools

p-curve	The closest relative: same input requirements, same significant-results-only design, same inability to handle non-significant results. p-curve diagnoses evidential value qualitatively and estimates average power; p-uniform estimates the bias-corrected effect size directly. Run both and compare.
PET-PEESE	Meta-regression approach that also produces a bias-corrected effect size estimate, but uses all studies (significant and non-significant) and models the relationship between effect size and standard error. PET-PEESE requires no distributional assumptions about the selection mechanism but is sensitive to heterogeneity in a different way. The two methods should converge in well-behaved literatures.
Trim-and-Fill	Non-parametric imputation approach that also produces a bias-corrected pooled estimate. Less theoretically principled than p-uniform but widely reported and easily understood. Convergence of trim-and-fill, PET-PEESE, and p-uniform on similar estimates substantially strengthens confidence in the bias-corrected value.
z-curve 2.0	Extends the p-value distribution approach to include non-significant results, estimating expected replication and discovery rates. Complements p-uniform by providing a field-level summary that does not require the significant-results-only assumption.
Excess Significance Test (TES)	Tests whether the observed count of significant results exceeds what would be expected given estimated study power. Complements p-uniform's within-distribution test with an across-literature count test.

Key Citations

- van Assen, M. A. L. M., van Aert, R. C. M., & Wicherts, J. M. (2015). Meta-analysis using effect size distributions of only statistically significant studies. *Psychological Methods*, 20(3), 293–309. <https://doi.org/10.1037/met0000025>
- van Aert, R. C. M. & van Assen, M. A. L. M. (2021). Correcting for publication bias in a meta-analysis with the p-uniform* method. *OSF Preprints*. <https://doi.org/10.31219/osf.io/zqjr9>
- van Aert, R. C. M. (2022). *puniform: Meta-analysis methods correcting for publication bias*. R package version 0.2.6. <https://cran.r-project.org/package=puniform>
- Carter, E. C., Kofler, L. M., Forster, D. E., & McCullough, M. E. (2015). A series of meta-analytic tests of the depletion effect: Self-control does not seem to rely on a limited resource. *Journal of Experimental Psychology: General*, 144(4), 796–815. <https://doi.org/10.1037/xge0000083>
- Simonsohn, U., Nelson, L. D., & Simmons, J. P. (2014). P-curve: A key to the file drawer. *Journal of Experimental Psychology: General*, 143(2), 534–547. <https://doi.org/10.1037/a0033242>
- Hagger, M. S., Chatzisarantis, N. L. D., Alberts, H., et al. (2016). A multilab preregistered replication of the ego-depletion effect. *Perspectives on Psychological Science*, 11(4), 546–573. <https://doi.org/10.1177/1745691616652873>